AN INVERTIBLE SHEAF OF NONPOSITIVE DEGREE ON EVERY COMPONENT OF A CONNECTED REDUCED CURVE HAS NO NONZERO SECTIONS UNLESS IT IS TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one cross-chapter label, recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Sign and triviality criteria for invertible sheaves on a proper curve. Stacks Project, Varieties, Lemma lemma-check-invertible-sheaf-trivial (tag 0B40). Turns a nonzero section on a component of nonpositive degree into a nowhere-vanishing one.

Source reference. Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k which is reduced, connected, and equidimensional of dimension 1. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. If $\deg(\mathcal{L}|_C) \leq 0$ for all irreducible components C of X, then either $H^0(X, \mathcal{L}) = 0$ or $\mathcal{L} \cong \mathcal{O}_X$.*

Proof. Let $s \in H^0(X, \mathcal{L})$ be nonzero. Since X is reduced there exists an irreducible component C of X with $s|_C \neq 0$. But if $s|_C$ is nonzero, then s is nonwhere vanishing on C by Lemma 0B40. This in turn implies s is nowhere vanishing on every irreducible component of X meeting C. Since X is connected, we conclude that s vanishes nowhere and the lemma follows. □

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